HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and applied engineering principles to design and test air-cooled chiller components, aligning with European market requirements and general supervision guidelines.
- Authored detailed product specifications and coordinated documentation release cycles to support new product development initiatives and existing product lines.
- Selected and integrated mechanical components, including heat exchangers, fans, compressors, and refrigerant piping, balancing cost, performance, and reliability targets for chiller systems.
- Executed comprehensive test programs for mechanical systems and components; analyzed test data to troubleshoot and modify sub-components, ensuring performance validation.
- Collaborated with design and manufacturing teams to conceptualize system configurations and product layouts, optimizing for a balance of cost, performance, and reliability.
- Supported application engineers, factory operations, and product technical service teams by providing technical expertise and contributing to sustaining engineering efforts.
- Containerized Java Spring Boot microservices for a financial-services operations dashboard using **Docker** and deployed them to an **AWS EC2** environment managed via **Jenkins** CI/CD pipelines.

SKILLS

Engineering Principles – Fluid dynamics, Thermal principles, Refrigeration, Mechanical design, System configuration, Component selection

Codes & Directives – European mechanical codes, Safety codes, CE marking, Pressure Equipment Directive (PED), Machinery Directive, EN 378, EN 14511, EU F-Gas Regulation, ErP framework

Data Analysis & Tools – Data analysis, Large data processing, **Power BI**, **Tableau**, **DAX**, **Power Query**, R, Python (pandas, NumPy)

Software Development – **Java**, **Spring Boot**, **REST APIs**, **Python**, **SQL**, **PostgreSQL**, **MySQL**, **Oracle**, Git, Jenkins, Docker, AWS (EC2, S3, Lambda, CloudWatch)

Project Management – Product development, Design reviews, Documentation, Change management, Agile methodologies, Scrum, Reporting

PROJECTS

Air-Cooled Chiller Component Design and Testing

- Designed and documented novel heat exchanger configurations for enhanced thermal transfer in new air-cooled chiller models, improving efficiency by an estimated **12%**.
- Developed and executed a testing protocol for fan and compressor sub-systems, analyzing vibration and thermal data to identify and rectify **3 critical** performance anomalies.
- Collaborated with cross-functional teams on the selection of **valves and refrigerant piping** for optimal system performance and adherence to safety directives.

Automated Production Data Analysis Pipeline

- Engineered a **Python** data pipeline on **AWS Lambda** to ingest and process sensor readings from production chiller units, reducing manual data analysis time by **40%**.
- Developed **Power BI** dashboards to visualize key performance indicators (KPIs) and operational metrics derived from the processed data, enabling faster identification of performance trends for **product engineering teams**.
- Wrote **SQL** queries against a **PostgreSQL** database to extract and validate historical operational data for trend analysis and correlational studies.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)